

Chapter 13

Radar False Target Deception Jamming Suppression Based on Linear Canonical Domain Line Segment Detection



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13.1 Introduction

The interrupted sampling repeater jamming (ISRJ) is a coherent deception jamming technique used in radar electronic warfare that poses a significant threat to radar systems [1, 2]. With the advancement of digital radio frequency memory (DRFM) technology, jamming systems can intercept emitted signals more rapidly and generate composite jamming by adjusting the jamming parameters [3]. The energy and frequency characteristics of the jamming signals can closely resemble those of the target, making it difficult for traditional methods to effectively counter the jamming. Therefore, accurately distinguishing between jamming and target signals, as well as suppressing the jamming, is crucial to ensuring the radar's functionality in complex electromagnetic environments and enhancing detection capabilities.

Existing anti-ISRJ methods include waveform design combined with mismatch filtering [3, 4], parameter estimation [5], and deep learning [6], all of which have produced many feasible solutions. Time-frequency (TF) analysis [7, 8], an important branch of signal processing, has also gained widespread attention in the context of anti-jamming research, both domestically and internationally. In [9], the method first applies a short-time Fourier transform (STFT) to the echo signal, then calculates waveform entropy and variance features to extract the target echo signal. In [10], a TF energy distribution-based method is used to precisely extract the jamming-free segments from the de-chirped signal, thereby generating a filter to suppress the jamming.

While there has been progress in recognizing compound jamming, relevant anti-jamming methods remain relatively scarce. Current anti-jamming techniques

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mainly focus on single-target and single-jamming scenarios, where the frequency differences between the signals are large and nonoverlapping, making it easier to distinguish between them. However, in multi-target scenarios, the jammer can alter parameters, causing the radar receiver to detect multiple jamming signals. Moreover, when the jamming energy is close to or even lower than that of the target signal, energy-based filtering methods may mistakenly classify the target as jamming and filter it out.

To address these issues, this chapter proposes a novel algorithm based on short-time linear canonical transform (STLCT) [11] and Hough line detection. The linear canonical transform (LCT) [12] is introduced into radar anti-jamming techniques, and STLCT is used to convert target and jamming signals into distinct long straight lines and discontinuous segments. Hough transform is then employed to detect the discontinuous segments, and filtering is applied based on their position to suppress the jamming.

The main contributions and innovations of our work are as follows:

- (a) For the first time, ISRJ suppression is performed in the LCT domain. Compared to traditional TF analysis methods like STFT, STLCT is more suitable for practical, complex electronic warfare environments. The flexibility of the three free parameters in STLCT provides superior TF resolution, enabling better differentiation between target and jamming signals. Additionally, a fast algorithm for STLCT (STFLCT) is developed to meet real-time processing requirements, making the method more applicable to practical situations.
- (b) The method does not rely on energy distribution, addressing the shortcomings of traditional methods in handling jamming with similar or low energy levels. By applying Canny edge detection and morphological operations such as dilation, the Hough detection results can more comprehensively cover the jamming regions, thus maximizing the signal-to-jamming ratio.

13.2 Preliminaries

13.2.1 Signal Model

The target echo signal can be expressed as

$$S(t) = \text{rect}\left(\frac{t}{T}\right) \exp\left(j2\pi\left(f_c t + \frac{1}{2}kt^2\right)\right) \quad (13.1)$$

where f_c is the carrier frequency, k is the frequency modulation slope, T is the pulse width, $\text{rect}\left(\frac{t}{T}\right)$ is a rectangular function.

The jammer intercepts the radar transmission signal, performs intermittent sampling processing on it, and then delays the transmission. The jamming signal model is [1]

$$J(t) = \sum_{n=1}^N \text{rect} \left(\frac{t - (n-1)T_s}{T_J} \right) \exp \left(j2\pi \left(f_c t + \frac{1}{2} k t^2 \right) \right) \quad (13.2)$$

where N represents the number of samples, T_J represents the slice width, T_s represents the sampling period.

The radar receiving signal is

$$r(t) = A_1 S(t - \tau_1) + A_2 S(t - \tau_2) + A_{j1} J(t - \tau_{j1}) + A_{j2} J(t - \tau_{j2}) + n(t) \quad (13.3)$$

where $n(t)$ is a Gaussian white noise, and $\tau_1, \tau_2, \tau_{j1}, \tau_{j2}$ denote the time delays of two target echo signals and two ISRJ signals, respectively.

13.2.2 Short-Time Linear Canonical Transform

As a generalization of the Fourier transform (FT) and the fractional Fourier transform (FRFT), the LCT is a useful TF analysis tool. Assuming that LCT of the input signal $x(t)$ is $L_A(x)(u)$ with parameter $A = (a, b, c, d)$, it is defined as [13]

$$L_A(x)(u) = \begin{cases} \int_{\mathbb{R}} x(t) K_A(t, u) dt, & b \neq 0 \\ \sqrt{d} e^{j \frac{cd}{2} u^2}, & b = 0 \end{cases} \quad (13.4)$$

where $a, b, c, d \in \mathbb{R}$ satisfying $ad - bc = 1$, and the kernel of LCT is

$$K_A(t, u) = \frac{1}{\sqrt{j2\pi b}} e^{j \frac{d}{2b} u^2 - j \frac{1}{b} t u + j \frac{a}{2b} t^2}, \quad b \neq 0 \quad (13.5)$$

In general, we only consider $b \neq 0$, because the LCT with $b = 0$ is only a chirp multiplication operation. The parameter of inverse LCT is $A^{-1} = (d, -b, -c, a)$, expressed as

$$x(t) = \int_{\mathbb{R}} L_A(x)(u) K_{A^{-1}}(u, t) du = \int_{\mathbb{R}} L_A(x)(u) K_A^*(u, t) du \quad (13.6)$$

where $*$ is the complex conjugate.

The definition of STLCT can be written as [11]

$$\text{STLCT}_f^A(t, u) = e^{-j \frac{t u}{b}} \int_{\mathbb{R}} f(\tau) e^{j \frac{a}{2b} \tau^2} \cdot g^*(\tau - t) e^{-j \frac{\tau^2}{2b}} d\tau \quad (13.7)$$

According to the definition (13.7) of STLCT, its calculation process can be decomposed into the following three steps

1. Modulating the input signal with a chirp function

$$f(t) \rightarrow f'(t) = f(t)e^{j\frac{a}{2b}t^2} \quad (13.8)$$

2. Performing STFT (with its argument scaled by $\frac{1}{b}$)

$$f'(t) \rightarrow \text{STFT}_{f'}\left(t, \frac{u}{b}\right) \quad (13.9)$$

3. Another modulation with a chirp signal

$$\text{STFT}_{f'}\left(t, \frac{u}{b}\right) \rightarrow \text{STLCT}_f^A(t, u) = e^{-j\frac{tu}{b}} \text{STFT}_{f'}\left(t, \frac{u}{b}\right) \quad (13.10)$$

13.3 Jamming Suppression

13.3.1 STLCT Domain Characteristics of Signals

This partly deduces the distribution characteristics of the LFM signal in the STLCT domain. Substitute $f(\tau)$ into the STLCT definition

$$\text{STLCT}_f^A(t, u) = e^{-j\frac{at^2}{2b}} \int_{\mathbb{R}} A e^{j(2\pi f_0\tau + \pi\beta\tau^2)} \cdot e^{j\frac{a\tau^2}{2b}} \cdot g^*(\tau - t) \cdot e^{-j\frac{\tau u}{b}} d\tau \quad (13.11)$$

Combine the exponents in the exponential terms

$$j\left(2\pi f_0\tau + \pi\beta\tau^2\right) + j\frac{a\tau^2}{2b} - j\frac{\tau u}{b} \quad (13.12)$$

This can be simplified to

$$j\left[\left(\pi\beta + \frac{a}{2b}\right)\tau^2 + \left(2\pi f_0 - \frac{u}{b}\right)\tau\right] \quad (13.13)$$

Define

$$\alpha = \pi\beta + \frac{a}{2b}, \quad \gamma = 2\pi f_0 - \frac{u}{b} \quad (13.14)$$

Then the exponent becomes

$$j \left[\alpha \tau^2 + \gamma \tau \right] \quad (13.15)$$

The STLCT is now expressed as

$$\text{STLCT}_f^A(t, u) = e^{-j \frac{at^2}{2b}} \int_{\mathbb{R}} Ag^*(\tau - t) e^{j(\alpha \tau^2 + \gamma \tau)} d\tau \quad (13.16)$$

To simplify $\alpha \tau^2 + \gamma \tau$, we complete the square

$$\alpha \tau^2 + \gamma \tau = \alpha \left(\tau^2 + \frac{\gamma}{\alpha} \tau \right) \quad (13.17)$$

Completing the square

$$\tau^2 + \frac{\gamma}{\alpha} \tau = \left(\tau + \frac{\gamma}{2\alpha} \right)^2 - \left(\frac{\gamma}{2\alpha} \right)^2 \quad (13.18)$$

Thus

$$\alpha \tau^2 + \gamma \tau = \alpha \left(\left(\tau + \frac{\gamma}{2\alpha} \right)^2 - \left(\frac{\gamma}{2\alpha} \right)^2 \right) \quad (13.19)$$

Substituting back

$$\text{STLCT}_f^A(t, u) = e^{-j \frac{at^2}{2b}} e^{-j \alpha \left(\frac{\gamma}{2\alpha} \right)^2} \int_{\mathbb{R}} Ag^*(\tau - t) e^{j \alpha \left(\tau + \frac{\gamma}{2\alpha} \right)^2} d\tau \quad (13.20)$$

Let

$$\xi = \tau + \frac{\gamma}{2\alpha}, \quad \tau = \xi - \frac{\gamma}{2\alpha} \quad (13.21)$$

The integral becomes

$$\text{STLCT}_f^A(t, u) = e^{-j \frac{at^2}{2b}} e^{-j \alpha \left(\frac{\gamma}{2\alpha} \right)^2} \int_{\mathbb{R}} Ag^*\left(\xi - t - \frac{\gamma}{2\alpha}\right) e^{j \alpha \xi^2} d\xi \quad (13.22)$$

Assume the window function is a Hamming window

$$g(\tau) = 0.54 + 0.46 \cos\left(\frac{2\pi \tau}{T}\right), \quad |\tau| \leq \frac{T}{2} \quad (13.23)$$

The conjugate is

$$g^*(\tau - t) = 0.54 + 0.46 \cos\left(\frac{2\pi(\tau - t)}{T}\right) \quad (13.24)$$

Substitute into the STLCT

$$\begin{aligned} \text{STLCT}_f^A(t, u) &= e^{-j\frac{at^2}{2b}} e^{-j\alpha(\frac{\gamma}{2\alpha})^2} \\ &\times \int_{\mathbb{R}} A \left[0.54 + 0.46 \cos\left(\frac{2\pi(\xi - t - \frac{\gamma}{2\alpha})}{T}\right) \right] e^{j\alpha\xi^2} d\xi \end{aligned} \quad (13.25)$$

Split the integral into two parts

$$\begin{aligned} \text{STLCT}_f^A(t, u) &= e^{-j\frac{at^2}{2b}} e^{-j\alpha(\frac{\gamma}{2\alpha})^2} \\ &\times \left[0.54 \int_{\mathbb{R}} A e^{j\alpha\xi^2} d\xi + 0.46 \int_{\mathbb{R}} A \cos\left(\frac{2\pi(\xi - t - \frac{\gamma}{2\alpha})}{T}\right) e^{j\alpha\xi^2} d\xi \right] \end{aligned} \quad (13.26)$$

The first integral is

$$\int_{\mathbb{R}} A e^{j\alpha\xi^2} d\xi \quad (13.27)$$

This is a standard Gaussian integral. The result is

$$\int_{\mathbb{R}} e^{j\alpha\xi^2} d\xi = \sqrt{\frac{\pi}{j\alpha}} = \sqrt{\frac{\pi}{\alpha}} e^{-j\frac{\pi}{4}} \quad (13.28)$$

Thus

$$0.54 \int_{\mathbb{R}} A e^{j\alpha\xi^2} d\xi = 0.54 A \sqrt{\frac{\pi}{\alpha}} e^{-j\frac{\pi}{4}} \quad (13.29)$$

The second integral is

$$\int_{\mathbb{R}} A \cos\left(\frac{2\pi(\xi - t - \frac{\gamma}{2\alpha})}{T}\right) e^{j\alpha\xi^2} d\xi \quad (13.30)$$

Using the Euler formula for cosine

$$\cos(x) = \frac{1}{2} [e^{jx} + e^{-jx}] \quad (13.31)$$

The integral becomes two parts

$$\frac{1}{2} \int_{\mathbb{R}} A e^{j\alpha\xi^2} e^{j\frac{2\pi(\xi-t-\frac{\gamma}{2\alpha})}{T}} d\xi + \frac{1}{2} \int_{\mathbb{R}} A e^{j\alpha\xi^2} e^{-j\frac{2\pi(\xi-t-\frac{\gamma}{2\alpha})}{T}} d\xi \quad (13.32)$$

Using Gaussian integral methods, both parts can be computed similarly to the first integral, and the final result can be expressed compactly.

Finally, the STLCT for the LFM signal with a Hamming window can be expressed as

$$\text{STLCT}_f^A(t, u) = e^{-j\frac{at^2}{2b}} \cdot e^{-j\alpha(\frac{\gamma}{2\alpha})^2} \cdot \left[0.54\sqrt{\frac{\pi}{\alpha}} + 0.46\sqrt{\frac{\pi}{\alpha}} \cdot \cos\left(\frac{2\pi}{T}\left(t + \frac{\gamma}{2\alpha}\right)\right) \right] \quad (13.33)$$

Among them, the parameter $\alpha = \pi\beta + \frac{a}{2b}$ will affect the frequency resolution. In addition, unlike STFT, the time-frequency slope of the LFM signal is equal to $b\beta$, which provides greater flexibility for adjusting the shape of the time-frequency distribution. These conclusions provide a reference for how to select appropriate a and b parameters in subsequent experiments.

13.3.2 Proposed Algorithm

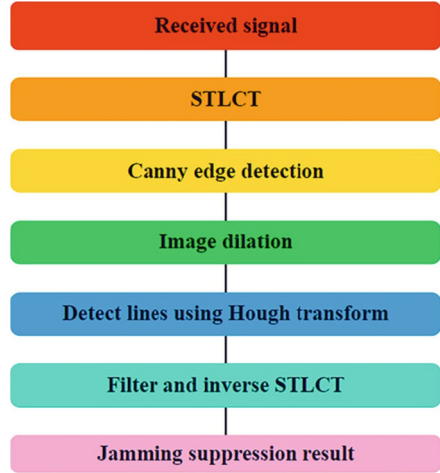
The flowchart of the proposed algorithm is shown in Fig. 13.1. The proposed method begins by performing a TF transformation on the received signal. Considering the selection of the free parameters in the STLCT, a rough range is first determined based on the conclusions derived above. Then, an optimal solution is obtained by traversing within this range to achieve the best separation between the target signal and jamming. To enhance practical applicability and enable faster implementation, a fast algorithm was specifically designed.

LCT can be decomposed into a cascade combination of basic LCT blocks through matrix decomposition [14]. Since each stage can be calculated in time, the overall LCT can also be. A systematic and detailed analysis of all possible decompositions of any LCT is carried out, which is decomposed into three basic operations: scaling, linear frequency modulation multiplication (CM), and Fourier transform (FT). Possible decompositions are considered, including three, four, and five cascade blocks.

After research, the STLCT calculation can be divided into three stages. Different from the traditional STLCT, by adjusting the order of operations, the operation cascade blocks are shown in formula (13.34), which reduces the total amount of STLCT calculation and improves the operation speed.

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{\nu}{\mu^2} & 1 \end{bmatrix} \begin{bmatrix} \mu & 0 \\ 0 & \frac{1}{\mu} \end{bmatrix} \quad (13.34)$$

Fig. 13.1 Algorithm flowchart



The first and third lower triangular matrices represent CM, the second matrix block represents FT, and the last is scaling. Therefore, STLCT can be completed in the following steps:

1. Modulating the input signal with a chirp function,
2. Performing STFT.
3. Another modulation with a chirp signal,
4. Scaled by its argument $\frac{1}{\mu}$.

After obtaining the TF image, the image is subjected to canny edge detection and morphological processing. The Canny edge detection algorithm [15] is used to extract edges from the input image. To reduce noise, the input image $I(x, y)$ is smoothed using a Gaussian filter:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}, \quad I_G(x, y) = I(x, y) * G(x, y)$$

The gradient magnitude G and direction θ are computed as:

$$G = \sqrt{G_x^2 + G_y^2}, \quad \theta = \arctan\left(\frac{G_y}{G_x}\right) \quad (13.35)$$

Non-maximum suppression is applied to thin the edges, and double thresholding is used to classify edges as strong or weak. The result of the Canny edge detection is a binary edge map highlighting the main structures in the image. For a binary image, the dilation of a set A (representing the foreground pixels) by a structuring element B is defined as [16]

$$A \oplus B = \{z \mid (B)_z \cap A \neq \emptyset\} \quad (13.36)$$

where A is the set of foreground pixels in the input binary image, B is the structuring element, $(B)_z$ represents the structuring element B translated by a vector z , $\cap$ represents the intersection operation, $\neq \emptyset$ means the intersection is not empty.

This definition states that a pixel z belongs to the result of dilation if the structuring element B , centered at z , overlaps with at least one foreground pixel in A . To detect straight lines in the edge map obtained from the previous step, the Hough transform [17] is applied. A line in the image is parameterized in polar form as

$$\rho = x \cos \theta + y \sin \theta \quad (13.37)$$

where ρ is the perpendicular distance from the origin, and θ is the angle of the perpendicular with the x -axis. For each edge pixel (x, y) , possible line parameters (ρ, θ) are computed and stored in an accumulator array

$$A(\rho, \theta) = A(\rho, \theta) + 1 \quad (13.38)$$

The peaks in the accumulator array correspond to the most likely lines in the image, with parameters (ρ, θ) indicating the detected lines.

13.4 Numerical Experiments

The experimental simulation setup consists of a received pulse signal, consisting of two LFM target echoes, two ISRJs, and noise. The time domain and frequency domain waveforms of the received signal are shown in Fig. 13.2. It can be seen that the target and the jamming are aliased together in the time domain and frequency domain and cannot be distinguished.

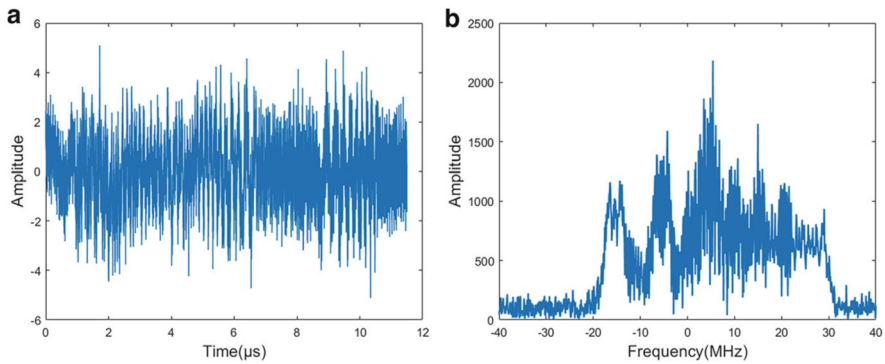


Fig. 13.2 Received signal. (a) Time domain and (b) frequency domain

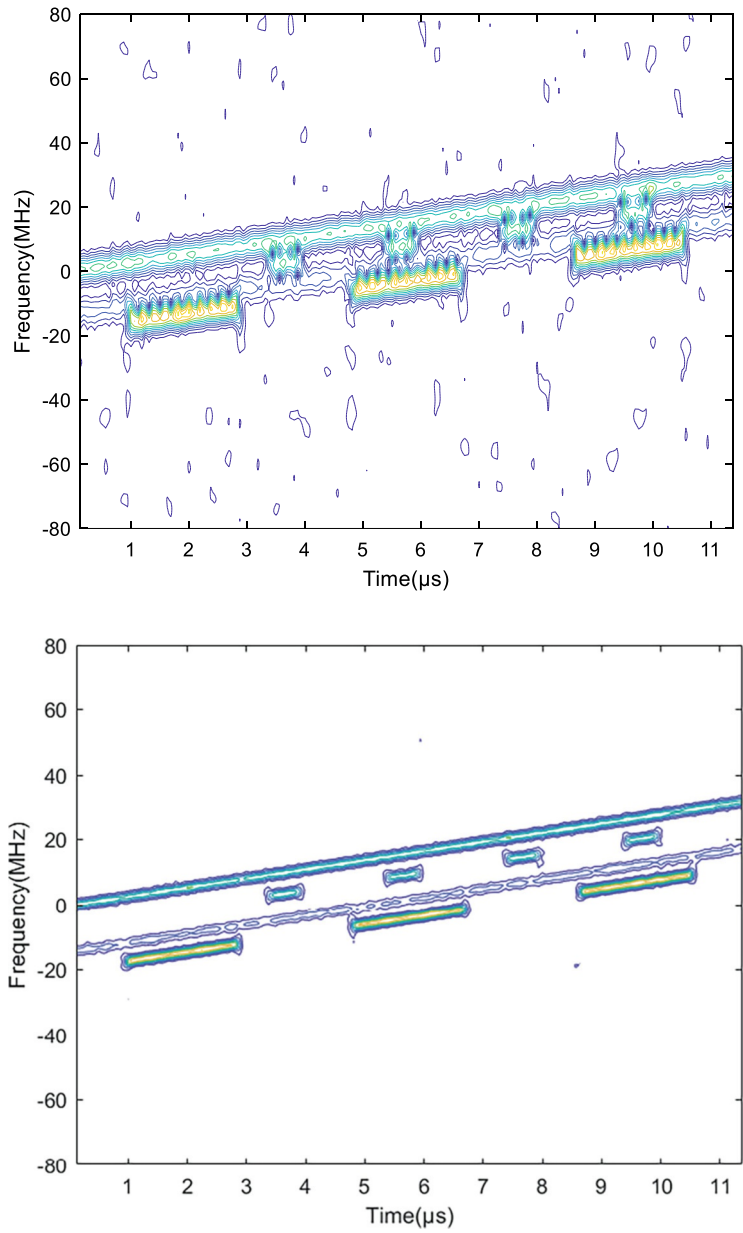


Fig. 13.3 Time-frequency image. (a) STFT and (b) STLCT

The signal is converted to the time-frequency domain through time-frequency analysis. Figure 13.3a is the result of STFT. It can be seen that due to the low time-frequency resolution of STFT, the target and the jamming are aliased and cannot be

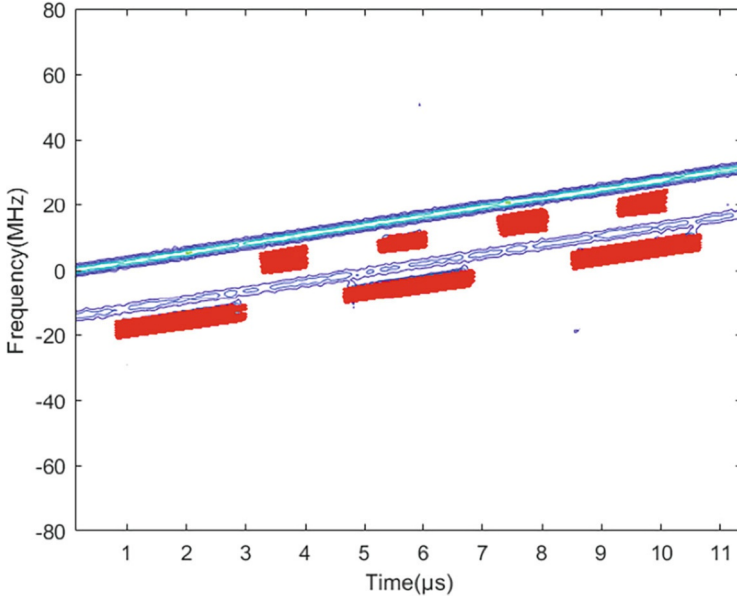


Fig. 13.4 Detection result

effectively distinguished. Figure 13.3b is the result of STLCT. Through the selection of free parameters, the target and the jamming are not aliased in the STLCT domain and can be distinguished.

Figure 13.4 is the detection result of Hough transform. It can be seen that the discontinuous line segments, that is, the jamming, are all detected, so that the filter can be set according to the detection position to filter and suppress the jamming. Finally, the STLCT inverse transform is performed on the filtering result to obtain the time domain signal after jamming suppression, and pulse compression [18] is performed to obtain the final output result, as shown in Fig. 13.5. Before jamming suppression, the false target caused by jamming is difficult to distinguish from the real target. After jamming suppression, the energy of the false target is greatly reduced, and the target signal is still retained, which significantly improves the radar's detection success rate for the target and reduces the false alarm probability.

13.5 Conclusion

In this chapter, a novel anti-jamming method for radar signal processing was proposed to effectively suppress ISRJ, a significant challenge in active deception jamming. By leveraging the LCT domain, we introduced a robust framework that combines TF analysis with line segment detection to enhance radar signal

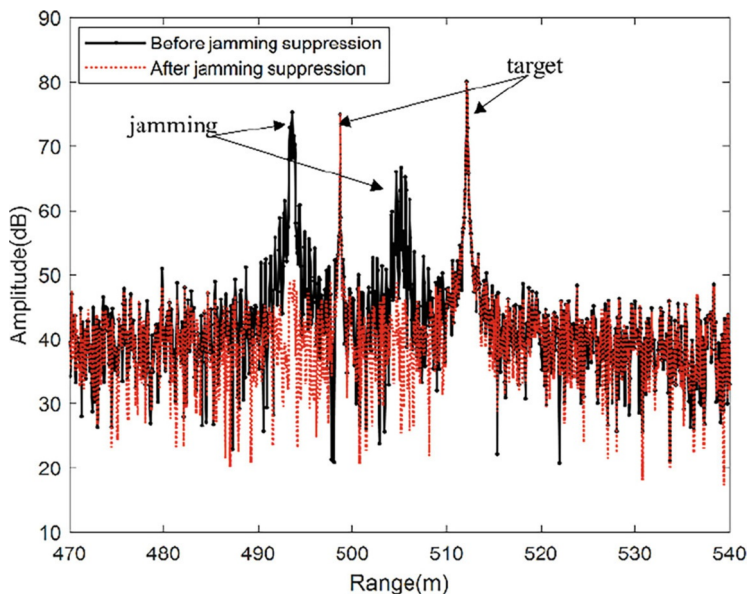


Fig. 13.5 Pulse compression output result

processing capabilities. The key contributions of this study include the derivation of signal distribution characteristics in the STLCT domain and the demonstration of how the flexibility of LCT's three free parameters can significantly improve TF resolution. A fast STLCT algorithm was designed to construct high-resolution TF images of radar echoes, enabling efficient and accurate representation of the signal. The proposed method further employed Canny edge detection and dilation operations to preprocess the TF images, followed by the detection of discontinuous line segments (representing jamming signals) using the Hough transform. These line segments were successfully identified and filtered out, and the suppressed signal was reconstructed using the inverse STLCT.

Simulation experiments validated the effectiveness of the proposed approach, demonstrating its capability to accurately suppress ISRJ and improve radar detection and identification performance. Future work could explore the integration of this method with other advanced signal processing techniques to further enhance its robustness against more complex jamming scenarios.

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